



Hafiz Muhammad Ali Zeeshan

Machine Learning | Generative AI | IoT & Emerging Technologies

🏠 Lahore, Pakistan



About me

Advancing next-generation innovation in Agentic AI, LLMs, and Machine Learning; skilled in leading teams to build high performance AI systems, optimize models, and deploy automation that drives business growth.

Areas of Interest

- ▶ AI Application Development
- ▶ AI Project Management
- ▶ Business Automation
- ▶ ML, Generative AI, & Agentic AI
- ▶ Algorithms & Optimizations
- ▶ Deep Reinforcement Learning
- ▶ Edge Computing & IoT Networks

AI Tools & Platforms

LangChain	██████████
Haystack	██████████
Hugging Face	██████████
AWS AI	██████████
Azure AI	██████████
Vertex AI	██████████
LlamaIndex	██████████
CrewAI	██████████
OpenAI API	██████████
Zapier/n8n	██████████

Certifications

AWS Certified AI Practitioner ✓
AI/ML, AWS AI services, AWS infrastructure, responsible AI practices. [View](#)

Databases and SQL for Data Science - IBM ✓
SQL, relational databases, Python integration. [View](#)

Deep Learning Specialization – Coursera ✓
CNNs, RNNs, Transformers, NLP, transfer learning. [View](#)

📁 EXPERIENCE

Jan-2025
Present

Head of AI | AI Strategy, Business Impact, Automation & Adv. AI Solutions
ADEPT TECH SOLUTIONS INC. CALIFORNIA, USA · Remote 📍

Lead AI strategy, design, and deployment. Build scalable multi-agent, MCP, RAG, and fine-tuned model systems optimised for cost and speed, deployed across cloud, hybrid, edge, and on-device with CI/CD. Own governance, ethics, and compliance; mentor AI teams and translate technical strategy into business impact.

Feb-2025
Present

AI Strategy Lead | Multi-Agent Architectures, RAG Pipelines, Cloud AI
AI WORKFORCE, UNITED KINGDOM · Remote 📍

Define and lead AI strategy across sales, marketing, and operations agents. Deploy LLM-powered multi-agent architectures and RAG pipelines over customer data and CRMs. Own model selection and evaluation, architect cloud-native AWS infrastructure, and oversee frontend, backend, and cloud engineering teams.

Jul-2018
Present

Founder & CEO | AI Services, Generative AI, MLOps & Automation
NEURALSTACKS, PAKISTAN · Remote 📍

Top Rated Plus freelancing agency offering diversified AI-based services across generative AI, machine learning, MLOps, and business automation.

Nov-2021
Present

AI Engineer | AI Development, Automation, and SaaS Solutions
IMPERIUM SOLUTIONS, USA · Part-time, Remote 📍

Designing and deploying AI-powered applications: chatbots, automation workflows, and SaaS platforms. Generative AI, NLP, RAG, multi-agent orchestration, MCP integration, and model fine-tuning, with cloud-native deployment (AWS, Azure, GCP) and MLOps pipelines.

Dec-2014
Present

Freelancer | Business Automation, SaaS, Agentic AI & LLM Solutions
UPWORK, USA · Remote 📍

100+ completed engagements. AI strategy and solution design, SaaS-based AI applications, generative AI, domain-specific fine-tuning, multi-agent orchestration, MCP integration, RAG, inference optimization, AWS deployments, process automation, and predictive analytics.

Nov-2023
Dec-2024

Principal AI Engineer | Project Management, Generative AI, QC
ADEPT TECH SOLUTIONS INC. CALIFORNIA, USA · Remote 📍

Built the AI delivery practice that the Head of AI role now leads: scalable multi-agent, MCP, RAG, and fine-tuned model systems, with responsible-AI review and engineer mentoring.

Sep-2023
Dec-2023

Sr. Algorithm Engineer | Computer Vision, Generative AI & 3D Reconstruction
SIX ATOMIC, SINGAPORE · Remote 📍

Designed and optimised algorithms for computer vision and generative AI, including image and video understanding, 3D reconstruction, object detection, and multimodal synthesis. Built and fine-tuned CNNs, transformers, and diffusion models for real-time, high-accuracy performance, and implemented model optimisation, quantization, and acceleration for cloud, edge, and embedded deployment.

Jul-2016
Dec-2021

Sr. Core Networks Engineer | Troubleshooting, Maintenance & Optimization
ZONG CMPAK LTD. FAISALABAD, PAKISTAN · Onsite 📍

Optimised national core network performance through analytics, capacity planning, and predictive maintenance. Daily maintenance of NSS elements (MSCs, MGWs, HLR, OMM); KPI monitoring and improvement; performance and fault reporting; LAC, GCI, and SAI configuration, including the Central-1 Huawei swap.

Jan-2015
Jul-2016

Core Network Engineer | Data Analytics, Network Efficiency & Resilience
ZONG CMPAK LTD. SAHIWAL, PAKISTAN · Onsite 📍

Maintained traffic channel availability above target, resolved service-affecting alarms, and optimised RAN performance. Traded off generator use against battery health to cut per-site cost.

Key AI Projects

Multi-Modal AI Content Creation & Marketing Automation

Built a content pipeline with GPT-4, Claude, Stable Diffusion, and ControlNet, integrated via Zapier, n8n or Make.com, delivering 400% more content at 60% lower cost. [View](#)

AI-Enhanced Real Estate Investment Platform

QGIS and satellite analysis with automated CMA, forecasting, and MLS integration for 300% faster opportunity detection and 40% higher accuracy. [View](#)

Intelligent Supply Chain Optimization & Demand Forecasting

LSTM/Transformer models with IoT and Monte Carlo simulations achieving 90%+ forecast accuracy and 35% lower inventory costs. [View](#)

Computer Vision Quality Control & Inspection System

YOLOv8 and Open3D defect detection (<100ms latency) with PLC integration, delivering 98%+ accuracy at 100+ items/min. [View](#)

AI-Driven Customer Support & Lead Generation Platform

GPT-4, LLaMA 3, and Whisper API for voice support with BANT qualification, boosting lead conversions by 40%. [View](#)

Enterprise AI-Powered Document Processing & Workflow Automation

GPT-4, Claude, TrOCR, and CrewAI with FastAPI/Make.com for 95%+ classification accuracy and GDPR/HIPAA compliance. [View](#)

AI Agents for Marketing Strategy Automation

LangChain and AutoGen agents for automated research, scheduling, and optimization via CrewAI. [View](#)

Healthcare AI Assistant & Patient Monitoring System

HIPAA-compliant assistant using TrOCR and FHIR, reducing admin work by 70% with real-time vitals monitoring and early alerts. [View](#)

Soft Skills

- Adaptability
- Critical Thinking
- Research Leadership

Languages

Urdu English

DEGREES

Jan-2024 Present	Ph.D. EE. (AI & Autonomous Systems) SEECs, NUST · Islamabad, Pakistan Areas of Research: Generative AI, Trustworthy LLMs, Multi-Agent AI Systems, Knowledge Distillation, Quantization, Inference Optimization, Next-Gen IoT & Cyber-Physical Systems, Edge Intelligence, and Intelligent 6G Communication Infrastructures	CGPA: 4.0/4.0
Sep-2021 Sep-2023	M.Sc. EE. (AI & AS · With Distinction) SEECs, NUST · Islamabad, Pakistan Majors: Machine Learning (ML), Artificial Neural Networks (ANN), Convex Optimization, Reinforcement Learning, Natural Language Processing, Mobile Robotics, Cloud Computing, and Advanced Wireless Communication	CGPA: 4.0/4.0
Nov-2010 Aug-2014	B.Sc. EE. (Electronics & Telecommunications) DEPARTMENT OF ELECTRICAL ENGINEERING, UET · Lahore, Pakistan Majors: Signals and Systems, Digital Communications, Wireless Communications, Data Structures, Control Systems, Integrated Electronics, Power Electronics and Embedded Systems	CGPA: 3.34/4.0

RESEARCH COLLABORATIONS

Sep-2025 Ongoing	NATO SPS Project – SHAILLA SCIENCE FOR PEACE & SECURITY PROGRAMME · Canada, Turkey, Pakistan Contributing to "Shielding HAPS Against Intelligent Link Level Attacks (SHAILLA)," focusing on AI-driven vulnerability analysis, intelligent attack modeling, LLM-based adversarial strategies, and autonomous countermeasure design. Project aims to evaluate E2E security of next-generation wireless systems using HAPS as base stations, combining agentic AI and ML for robust defense mechanisms.
Jul-2025 Ongoing	NUST–Huzhou University Joint Research Collaboration SEECs, NUST - SIE, Huzhou · Pakistan - China Collaborating with Huzhou University on ML-based wireless communication systems, integrating AI, LLMs, edge computing, and IoT for future-ready 6G networks.
Jan-2024 Ongoing	Information Processing and Transmission (IPT) LAB SEECs, NUST · Pakistan Collaborating under the supervision of Dr. Syed Ali Hassan on next-generation wireless systems, focusing on AI, LLMs, autonomous agents, edge computing, and Next-Gen IoT for resilient and intelligent 6G networks.

PUBLICATIONS

Oct-2025	LLM-Based Retrieval-Augmented Generation: A Novel Framework for Resource Optimization in 6G and Beyond Wireless Networks , <i>IEEE Communications Magazine</i> , Vol. 63, No. 10, pp. 60–67. Zeeshan, H.M.A., Umer, M., Akbar, M., Kaushik, A., Jamshed, M.A., Jung, H. and Hassan, S.A.
Sep-2025	Toward Intelligent and Sustainable IoT: A DRL Approach for Backscatter-Based Communication Under QoS Constraints , <i>IEEE Open Journal of the Communications Society</i> , Vol. 6, pp. 7589–7603. Zeeshan, H.M.A., Ullah, S.A., Solaija, M.S.J., Jamshed, M.A., Mojahedian, M.M., Jung, H. and Hassan, S.A.
Aug-2025	LLM-Enhanced Dynamic Spectrum Management for Integrated Non-Terrestrial and Terrestrial Networks: A Multi-Objective Optimization Approach , <i>Mobile Networks and Applications</i> , Vol. 30, No. 3–4, pp. 459–470. Zeeshan, H.M.A., Ahmad, A., Hassan, S.A., Solaija, M.S.J. and Chauhdary, S.H.
Apr-2025	RAG Powered LLMs for QA: Evolution, Challenges, Applications, and Future Directions , <i>ComTech 2025</i> , pp. 1–6. <i>Best Paper, Software Engineering Track</i> . Zeeshan, H.M.A., Faizan, M., Zia, U. and Gohar, A.
Dec-2023	DDPG-Based Sum Rate Optimization for Opportunistic Backscatter NOMA Networks , <i>IEEE GLOBECOM 2023</i> , pp. 3312–3317. Zeeshan, H.M.A., Ullah, S.A., Hassan, S.A., Ding, Z. and Jung, H.
Aug-2023	MS Thesis: Performance Analysis of CR-NOMA Enabled Backscatter Communications Using Deep Reinforcement Learning , <i>NUST</i> . Hafiz Muhammad Ali Zeeshan.
Sep-2017	Advanced Techniques for Control of Smart Grids , <i>IEEE Smart Cities Conference</i> . Zeeshan, H.M.A., Naeem, M. and Nisar, F.
May-2017	A Review of Distributed Control Techniques for Power Quality Improvement in Micro-Grids , <i>IOP Conf. Series: Materials Science and Engineering</i> , Vol. 199, No. 1. Zeeshan, H.M.A., Nisar, F. and Hassan, A.